

Nicholas Kashani Motlagh

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Education

Ohio State University

Ph.D. in Computer Science

Columbus, OH

Aug 2021 – Aug 2026 (conferral: Aug 9)

Dissertation: *Answering Under Uncertainty: Abstention, Ambiguity, and Recoverability* (defended July 8; requirements complete)

M.S. in Computer Science and Engineering

Aug 2021 – May 2025

Advisor: Prof. Jim Davis; Minors: Mathematics and High-Performance Computing

GPA: 3.70

Ohio State University

B.S. in Computer Science and Engineering with Honors

Columbus, OH

Aug 2017 – May 2021

Minor: Mathematics; Scholarships: Maximus, Ten-Hai Lai, Ansel, Name and Seal

GPA: 3.86

Experience

Graduate Research Associate, Computer Vision Lab

Aug 2021 – Jul 2026

Ohio State University, Columbus, OH

- **Retrieval-Augmented Selective QA** (anonymous manuscript in preparation, 2026): studies when retrieval-conditioned revision repairs or harms short-form answers to guide return, revise, or abstain decisions for a fixed stack.
- **Assessing the Role of Imagery in Multimodal Machine Translation** (WMT 2024): introduced ImageCoMMuTE metrics separating visual understanding from final translation decisions.
- **Naturally Constrained Reject Option Classification** (MVA 2025): developed Binomial-CDF reject-option constraints with per-class thresholds across synthetic, image, and text datasets.
- **Learning When to Say “I Don’t Know”** (ISVC 2022, Springer Best Paper Award): learned per-class softmax thresholds for abstention without user-defined rejection costs.

Graduate Teaching Associate / Course Grader, Ohio State University

Aug 2023 – Dec 2025

- Served as GTA through May 2025; graded machine learning and NLP coursework and led Excel-based computing labs.

Technical Analyst II

May 2025 – Present

DCS Corporation / Air Force Research Laboratory, Dayton, OH

- Conduct applied machine learning research and evaluation for government-sponsored programs.

Graduate Research Intern

Summers 2022–2024

Air Force Research Laboratory, Dayton, OH

- Summer 2024: Adapted JEPA and MAE transformers for multimodal EO/SAR representation learning in low-data regimes.
- Summer 2023: Developed Reject Option Beam Search to improve machine translation quality at large beam widths.
- Summer 2022: Built the end-to-end training procedure for Naturally Constrained Reject Option Classification.

Undergraduate Research Intern, Air Force Research Laboratory

Summers 2020–2021

- **Temporal Satellite Imagery** (ICCV Workshop 2021): built an OpenStreetMap-guided Python pipeline for remote-sensing change detection and classification datasets.

Selected Publications and Manuscripts

- [1] **N. Kashani Motlagh** et al. Anonymous manuscript in preparation on recoverability in retrieval-augmented short-form QA, 2026.
- [2] **N. Kashani Motlagh** et al. “Naturally Constrained Reject Option Classification.” *Machine Vision and Applications* 36, Article 9, 2025.
- [3] **N. Kashani Motlagh** et al. “Assessing the Role of Imagery in Multimodal Machine Translation.” *WMT*, pp. 1428–1439, 2024.
- [4] **N. Kashani Motlagh** et al. “Learning When to Say ‘I Don’t Know.’” *ISVC*, pp. 196–210, 2022. Springer Best Paper Award.
- [5] **N. Kashani Motlagh** et al. “A Framework for Semi-automatic Collection of Temporal Satellite Imagery for Analysis of Dynamic Regions.” *ICCV Workshop*, pp. 704–712, 2021.

Technical Skills

Languages and Tools: Python, PyTorch, vision-language models, Hugging Face, Git, Slurm, Singularity, LaTeX